TEXORA · LECTURE SHEET

Slub, Inject Slub, Neppy, Melange, Siro, Wrap Yarn and Compact Yarn

Seven commercially important special yarns, each created by a different trick: programmed drafting, injected fibres, fibre blending, doubled roving feed, or pneumatic condensing. This sheet gives the structure, production method, properties and uses of each, yarn by yarn.

YM-4103 · SPECIAL YARN MANUFACTURING · TOPIC 4 · LEVEL 4 TERM 1 · YARN MANUFACTURING SPECIALIZATION

1. Slub yarn

Structure. A single spun yarn of fine base count interrupted at programmed intervals by thicker, tapered sections (slubs) of higher count. The slub is the same fibre, spun in the same pass — fibre supply per unit length simply rises through the slub. **Production.** On a ring frame fitted with a slub device / servo-controlled drafting drive: the middle and back roller speeds are varied to a stored programme, momentarily reducing draft so a thicker length is delivered, then restoring base draft. Programmes set slub length (typically a few cm to tens of cm), slub thickness as a multiple of base count, spacing, and random variation so the pattern does not repeat mechanically in fabric. A multicount variant varies the base count itself in steps along the yarn. **Properties.** Strength is governed by the thin base sections, so mean count alone overstates usable strength; evenness CV is meaningless as a fault measure because variation is intentional — quality is judged against the slub programme. **Uses.** Denim, casual shirting, dress fabrics and furnishings seeking a hand-crafted, linen-like texture.

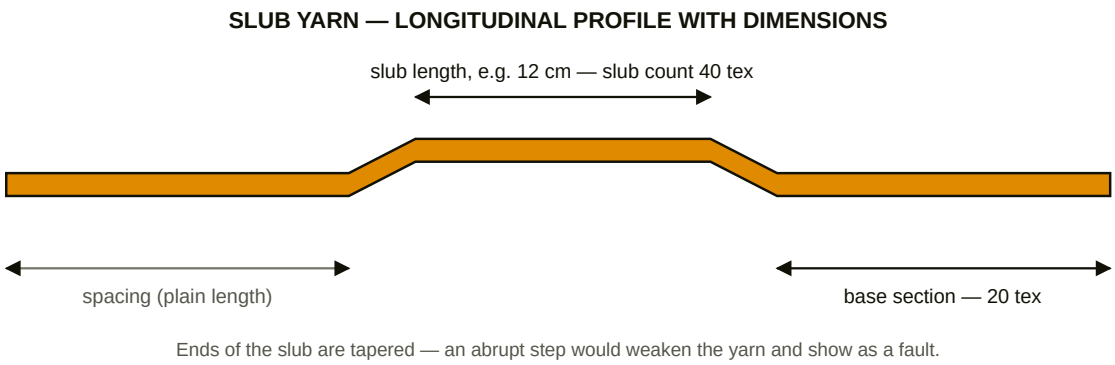


Figure 1 — Slub yarn profile. A tapered thick place of double the base count, separated by plain base-count sections; all three dimensions come from the slub programme.

WORKED EXAMPLE — AVERAGE COUNT OF A SLUB YARN

Base count 20 tex; slubs are 40 tex and occupy 10% of the yarn length, the remaining 90% being base yarn.

Average linear density = 0.90 × 20 + 0.10 × 40 = 18 + 4 = **22 tex**.

Lesson: a skein test on this yarn reports about 22 tex, not 20 tex. Fabric weight calculations must use the *average* count, while strength expectations must use the *base* count — using one number for both jobs is the classic slub-yarn costing error.

2. Inject slub yarn

Structure. A base yarn into which short tufts of a different fibre — usually a contrasting colour or a different material — are inserted at intervals, so coloured flecks or short slub-like tufts appear embedded in an otherwise normal yarn. **Production.** An injection device at the spinning frame feeds small, metered tufts of the effect fibre into the drafted strand just before twist insertion, at programmed intervals; the twist of the base yarn traps the tufts. Unlike a drafting slub, the effect is a *different material*, so colour contrast is sharp and the tuft can also differ in fineness or lustre. **Properties and uses.** Point colour effects without dyeing separate yarns; used in tweeds, casual knits and furnishing yarns. The quality risk is tuft loss if insertion depth or twist is insufficient, so injection timing relative to the twisting point is the critical setting.

3. Neppy yarn

Structure. A yarn carrying small, dense fibre entanglements — neps — as a deliberate surface effect. Nep here is a design feature, not the carding fault of the same name: the neps are produced or preserved on purpose, often in a contrasting dye affinity so that after piece dyeing the neps appear as pale or coloured specks (the familiar heathered denim and “nep denim” look uses neps that dye differently from the ground). **Production.** Routes include spinning from nep-rich prepared sliver, adding manufactured nep material during preparation, or using fibres of differing maturity/dye affinity so nep-like specks develop in dyeing. **Properties and uses.** Casual, rustic surface interest in denim, jersey and sweatshirt fabrics. Control is by nep size and frequency per metre against a reference sample — the same profile discipline as Topic 3 — because uncontrolled nep is indistinguishable from bad carding.

4. Melange yarn

Structure. A spun yarn made from a blend of fibres of two or more colours (or dye affinities) mixed *before* spinning, so every cross-section contains a mixture and the yarn shows an intimate, heathered colour rather than stripes. **Production.** Dyed fibres (typically stock-dyed) are blended at blowroom or

drawframe stage in exact percentage proportions; doubling at drawframe evens the blend along the length. The blend ratio is the colour: small ratio changes shift the shade visibly, so weighing accuracy at blending is colour quality control. **Properties and uses.** Soft, muted, premium-looking heather shades with excellent shade depth; widely used in knitwear, T-shirts and suiting. Advantages over yarn- or piece-dyeing: colourfast multi-tonal shades impossible to obtain by later dyeing, and no white-fleck risk from undyed fibre if blending is thorough. Faults: streakiness from poor blending, and shade drift between lots from ratio errors.

5. Siro yarn (Siro-spun)

Structure. Two rovings are fed side by side through one drafting system and converge at a small distance before the spindle, so the delivered yarn is effectively two fine strands twisted together at the moment of formation — a ply-like yarn produced in a single spinning pass. **Production.** A modified ring frame carries double roving guides and roving stop motions; the key variable is **strand spacing** — the distance between the two drafted strands at the front rollers. At the convergence point the strands wrap around each other, trapping surface fibres between them. **Properties.** Compared with a single yarn of the same resultant count, Siro yarn is smoother, less hairy, stronger in abrasion and more even-looking, at far lower cost than true two-ply yarn (which needs a separate twisting pass and two spinning passes at half count). **Uses.** Lightweight weaving yarns (its original purpose — a weavable “two-ply from one pass” for wool suiting), shirting and knitwear. A related variant, Siro-compact, adds the condensing system of Section 7.

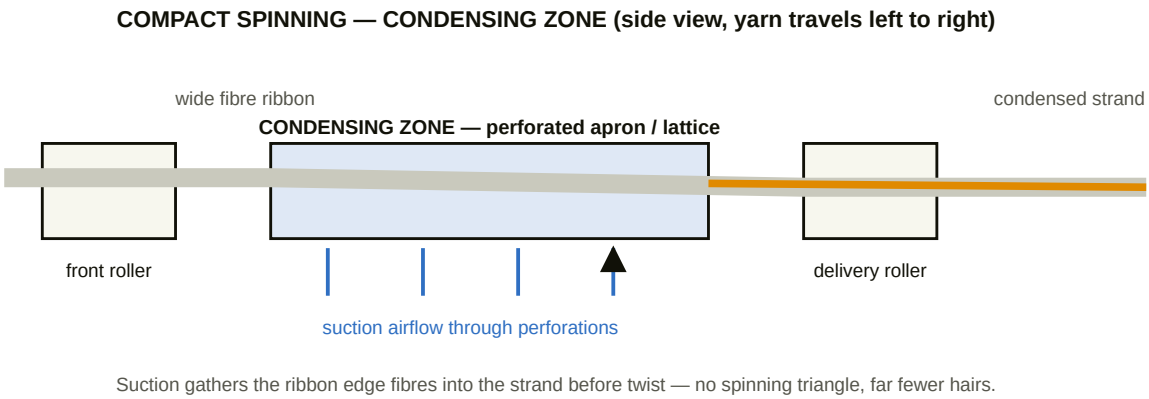


Figure 2 — Compact spinning condensing zone. A perforated apron over a suction slot condenses the drafted ribbon after the front roller; twist then runs into a narrow strand instead of a wide spinning triangle.

6. Wrap yarn

Structure. A core yarn (or bundle of parallel, untwisted fibres/filaments) held together solely by a fine yarn wrapped helically around it — the core itself carries no twist. **Production.** Hollow-spindle

machine (Topic 2): the core passes through the hollow spindle while the wrapping yarn, carried on the spindle package, winds around it once per revolution; a second spindle can add a counter-wrap for balance and security. Wraps per metre = spindle speed / delivery speed. **Properties.** Because the core is untwisted, wrap yarns are bulky and soft for their strength, and delicate or weak core materials (loose fibre bundles, textured yarns, even pre-made fancy yarns) can be converted into a usable yarn without twisting damage. The wrap is also the whole of its security: under-wrapped yarns slip and grin. **Uses.** Hand-knitting and furnishing yarns, fancy-yarn intermediates, and as the wrapping principle inside many core-spun and fancy constructions.

7. Compact yarn

Structure. A ring-spun yarn in which the drafted fibre ribbon is pneumatically condensed before twist insertion, so nearly every fibre — including edge fibres — is incorporated into the yarn body under tension. The spinning triangle of conventional ring spinning (where edge fibres escape twist and become hairs or fly) is virtually eliminated. **Production.** After the front roller the ribbon passes over a condensing zone: perforated apron, perforated drum or lattice apron over a suction slot connected to a vacuum source. Air flowing through the perforations draws edge fibres inward and rolls the ribbon into a narrow, dense strand; the delivery roller then passes it to twist insertion and winding (Figure 2). **Properties — typical reported ranges vs. conventional ring yarn of the same count and material:** hairiness (long hairs) reduced by roughly half or more (typical improvements of about 50–80% in hairs longer than 3 mm are commonly reported), tenacity higher by typically about 5–15%, evenness slightly better, and markedly less fly generation in weaving, with better dye appearance and pilling resistance. These are typical ranges, not guarantees — gains depend on fibre, count and machine condition. **Uses.** Premium shirting, fine knitwear and any product where low hairiness and a clean, lustrous surface justify the extra machine cost and suction energy.

YARN	DEFINING TRICK	SEPARATE BINDER/CORE?	BEST-KNOWN USE
Slub	Programmed draft variation	No	Slub denim, casual shirting
Inject slub	Tufts of foreign fibre injected pre-twist	Base yarn traps tufts	Tweed fleck effects
Neppy	Deliberate neps / differential dyeing	No	Nep denim, rustic jersey
Melange	Coloured fibres blended before spinning	No	Heather knitwear, T-shirts

Siro	Two rovings, one pass, strand spacing	Self-plying	Lightweight suiting, shirting
Wrap	Untwisted core + helical wrap	Wrap is the binder	Bulky knitting yarns
Compact	Suction condensing before twist	No	Premium low-hairiness shirting

Glossary

Slub programme — stored settings of slub length, thickness, spacing and variation. **Injection** — insertion of foreign fibre tufts into a drafted strand before twisting. **Nep** — small dense fibre entanglement; a fault in ordinary yarn, an effect here. **Melange** — intimate blend of differently coloured fibres spun into one heathered yarn. **Strand spacing** — distance between the two drafted strands in Siro spinning, controlling convergence. **Wrap yarn** — untwisted core held only by a helical wrapping yarn. **Condensing zone** — suction section that narrows the fibre ribbon in compact spinning. **Spinning triangle** — fibre fan between front-roller nip and twist point in ring spinning, source of edge-fibre loss.

Key points

• Slub = programmed thickness in one yarn; judge it against its programme, cost it by average count, rate its strength by base count.
• Inject slub inserts a different material; neppy yarn designs with entanglements; melange colours the fibre before spinning, not the yarn after.
• Siro spinning makes a ply-like yarn in one pass — strand spacing is its critical setting.
• Wrap yarn: the core has no twist at all; the helical wrap is structure, binder and security in one.
• Compact spinning deletes the spinning triangle by suction condensing — typical gains: long hairs down by about half or more, tenacity up by about 5–15%.

Practice

1. A slub yarn has base count 25 tex, slubs of 50 tex occupying 8% of its length. Calculate its average count.
2. A buyer wants a heather-grey T-shirt yarn whose shade must not change if the fabric is later garment-dyed a darker colour, and a second buyer wants sharp multicolour flecks in a tweed. Choose melange, inject slub or neppy construction for each and justify the choice in one sentence each.

3. Explain why a Siro yarn is less hairy than a single yarn of the same resultant count, referring to what happens at the convergence point.
4. State two distinct consumer-visible benefits of compact yarn in a finished shirt, and link each to the elimination of the spinning triangle.

Answer key

1. Average = 0.92 × 25 + 0.08 × 50 = 23 + 4 = **27 tex**.
2. T-shirt: **melange** — the colour lives in the fibres themselves as an intimate blend, giving a true heather that is a fibre property, not a surface dye pattern. Tweed flecks: **inject slub** — metered tufts of contrasting fibre give sharp, discrete colour points that blending (melange) would dissolve into an overall tint.
3. In Siro spinning the two strands converge and wrap around each other before final twist; surface fibres that would protrude from a single strand are trapped *between* the two strands at convergence and locked inside the yarn, so fewer fibre ends remain free to form hairs.
4. (i) Cleaner, sharper appearance and better print/dye definition — edge fibres are inside the yarn instead of forming a hairy halo that scatters light and dye. (ii) Less pilling and better abrasion life — with no spinning triangle, almost no short edge fibres escape twist to work out as pills or fly; nearly all fibres are bound under tension into the yarn body.
